European Commission

Talent for AI in Science

Mutual Learning Exercise on National Policies on AI in Science

Third thematic report

PSF CHALLENGE

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Report



Research and Innovation

Talent for AI in Science. Mutual Learning Exercise on National Policies on AI in Science

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EUROPEAN COMMISSION

Talent for AI in Science

Mutual Learning Exercise on National Policies on AI in Science

Third thematic report

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2025

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1. Introduction

This thematic report is the second in a series within the Mutual Learning Exercise (MLE) on National Policies for Artificial Intelligence (AI) in Science, focused on the topic of Talent for AI in Science. The purpose of this MLE is to exchange information and experiences as well as identify good practices to support national policies for AI in Science. This report provides an overview of the state of play and main policy challenges in relation to Talent for AI in Science in the countries taking part in the MLE and policy recommendations. It is based on desk research, a survey on talent for AI in science among participating MLE countries, and discussions at the second country visit in Oslo between 6 and 7 February 2025.

1.1. Background

1.1.1. AI in Science

AI in science refers to using artificial intelligence to improve and speed up research across various disciplines. This includes everything from traditional machine learning techniques and large language models used by individual researchers, to more advanced systems that can conduct experiments and analyse data on their own. Recent AI advances are changing the way researchers work in fields such as materials science, drug design, quantum physics, and medicine, making research more efficient and better equipped to tackle challenges like climate change and pandemics.

The role of AI in research is gaining wider recognition, as reflected in the 2024 Nobel Prizes awarded for breakthroughs in neural networks and for applying AI to protein structure prediction. These honours highlight how AI is becoming an integral tool in modern science, while also pointing to the ongoing questions about its future capabilities and applications.

1.1.2. Talent for AI in Science

Talent is the cornerstone of any scientific advancement, and when it comes to fast-developing fields such as AI, it is even more critical. AI is rapidly transforming how science is conducted, enabling faster data analysis, improved predictive modelling, and new methods of discovery. However, these advancements hinge on the availability of skilled individuals who can harness AI effectively. The ability to attract, nurture, and retain such talent determines whether a country or institution can fully realise the potential of AI in science.

At its leading edge, AI-driven science requires a unique blend of expertise in fields like computer science, mathematics, statistics, and domain-specific knowledge. This makes the cultivation of talent in AI in science particularly challenging. Individuals with the ability to merge deep technical AI expertise with insights into complex scientific problems are rare but vital. These professionals not only drive innovation but also help bridge the gap between theoretical AI capabilities and practical scientific applications.

The global competition for AI talent is intensifying[1]. With industries offering lucrative opportunities, many researchers are drawn away from academia and public institutions, creating a talent drain in the scientific sector[2]. Recent evidence suggests that this drain often involves the transition of top

[1] https://www.bcg.com/publications/2024/whos-winning-race-stem-ai-talent?linkId=661900158

[2] Ahmed, N., Wahed, M., & Thompson, N. C. (2023). The growing influence of industry in AI research. Science, 379(6635), 884-886.

scientists from leading universities and has unintended negative effects on scientific productivity[3]. The Global AI Talent Tracker 2.0[4] has been tracking top AI talent's movements from where they studied to where they are working now. Figure 1 shows the flow of top AI talent who finished undergraduate studies in Europe. We can see a clear outflow of talent from Europe mainly to the USA, followed by China, Australia, Canada, the UK, and other countries. However, there is some inflow from undergraduate to graduate school, where talent originates mainly from China, followed by India, the USA, Russia, and other countries. However, it is clear that Europe has problems with retaining talent also after graduate school (see Figure 2), with many students leaving Europe to pursue post-grad work in the USA, China, the UK, India, and other countries.

While this data offers valuable insights, it is important to note some limitations. The Global AI Talent Tracker relies solely on the data about authors who have published a paper at NeurIPS, a major but highly specialised conference focused on deep learning and neural networks. As a result, it may not capture talent working in other key areas of AI who publish at different venues, and therefore may not fully reflect the broader landscape of global AI research[5].

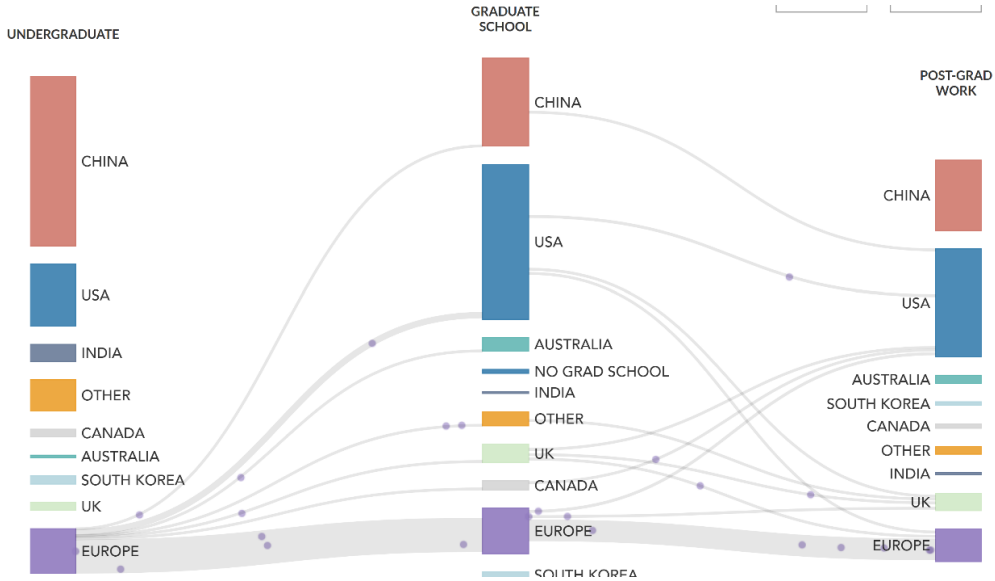


Figure 1: Flow of top AI talent who finished undergraduate studies in Europe.
Source: Marcopolo.org

[3] Jurowetzki, R., Hain, D.S., Wirtz, K. et al. The private sector is hoarding AI researchers: what implications for science?. *AI & Society* (2025); Gofman, M., & Jin, Z. (2024). Artificial intelligence, education, and entrepreneurship. *The Journal of Finance*, *79*(1), 631-667.

[4] https://macropolo.org/interactive/digital-projects/the-global-ai-talent-tracker/

[5] NeurIPS, while considered one of the top AI conferences, focuses primarily on deep learning and neural networks. Other important areas of AI research—such as symbolic reasoning, reinforcement learning, robotics, and AI ethics—are often represented at different conferences like AAAI, IJCAI, or ICML. Limiting the sample to NeurIPS authors may therefore skew the representation of global AI talent.

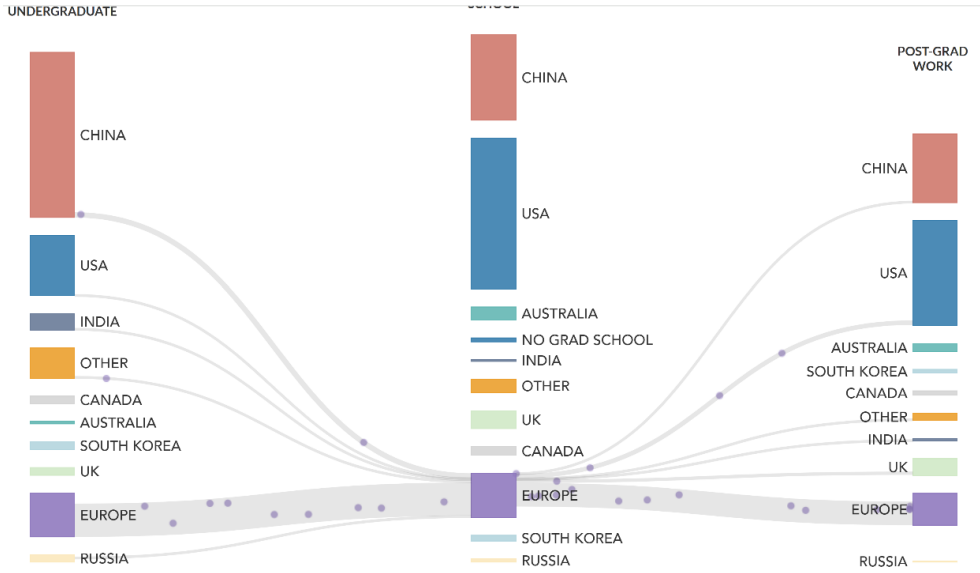


Figure 2: Flow of top AI talent coming to graduate school in Europe and after their graduation.
Source: Marcopolo.org

Addressing this challenge is crucial for ensuring that public and academic research environments remain vibrant and capable of leveraging AI for societal benefit.

Talent for AI in science is more than a resource, it is a strategic necessity. It enables innovation, strengthens global competitiveness, and empowers science to tackle pressing challenges, from climate change to disease eradication. Without robust strategies to develop and retain such talent, the transformative potential of AI in science risks being underutilised.

Across the surveyed MLE countries, the **greatest barriers to developing AI talent for science** include inadequate funding for AI research and limited access to specialised training programmes for scientists. Many respondents noted that insufficient educational courses focusing on AI and a lack of interdisciplinary collaboration further restrict opportunities to develop the necessary skills. Competition with the private sector was frequently mentioned, as more attractive salaries and favourable working conditions in industry often draw talent away from academic and public research roles. In addition, some countries pointed to challenges such as the limited availability of state-of-the-art computational facilities and overall AI computing infrastructure, which further hinder the effective development of AI expertise.

1.2. Scope and purpose of the thematic report

This report looks at the important role of diverse, skilled individuals in applying AI to science. It encompasses not only dedicated AI researchers but also domain researchers in need of upskilling, experts who bridge AI and specific scientific fields (i.e. hybrid roles), and support staff. The report aims to provide a clear picture of the current situation and the main challenges in attracting, training, and retraining this broad spectrum of AI talent within the EU. By sharing insights and experiences, we seek to propose ways to better develop, support, and sustain these various talent groups – ensuring that policies address the full diversity of expertise necessary for scientific innovation in AI for science.

The report covers key areas such as attracting and keeping skilled people, providing training, encouraging teamwork across different fields, building partnerships between public and private

organisations, and improving AI education. These areas are crucial for creating a strong pool of talent and helping AI make a real difference in scientific research.

As AI technologies continue to evolve, including advancements in generative AI (GenAI) and large language models (LLMs), the skills required to effectively use AI in scientific research are also changing. The increasing accessibility of AI tools raises questions about whether future scientists will need deep expertise in machine learning or if AI literacy and prompt engineering will suffice for many applications. This rapid transformation necessitates an adaptable and forward-looking educational approach to ensure researchers remain equipped with relevant skills. Therefore, a key policy consideration is the **need for agile, responsive education systems that can continuously adapt to emerging AI capabilities and research needs.**

The purpose of this report is to help policymakers, researchers, and institutions think about and act on these challenges. It aims to support strategies that ensure AI talent grows and thrives, so science can continue to tackle big challenges and benefit society as a whole.

1.3. The survey

To gain an indicative understanding of the state of talent for AI in science in the countries participating in the MLE, a survey was conducted and distributed among the participating MLE countries. The survey, combined with that for Topic 1 (Framework conditions and funding for AI in Science), comprised two parts: the first was directed at MLE participants and policymakers, and the second targeted at AI scientists from the MLE countries.

Eleven responses were received from participants from Austria, Flanders (Belgium), Estonia, Germany, Ireland, Italy, the Netherlands, Norway, and Slovenia, noting that some countries provided more than one response. The second part, aimed at AI scientists, yielded 16 responses from Austria, Flanders (Belgium), Estonia, Germany, Ireland, Italy, the Netherlands, and Slovenia. While these responses offer valuable insights, it is important to note that they are neither comprehensive nor representative of all EU countries or the entire scientific community. The limited sample size, combined with the possibility of self-selection bias, means some perspectives may be underrepresented. As such, the findings should be viewed as indicative rather than definitive and are intended primarily to stimulate further discussion and inquiry.

The majority of AI scientists from the surveyed MLE countries view interdisciplinary research involving AI as crucial for the future of science. Life sciences (biology and medicine) and engineering emerge as the most commonly integrated fields, with many participants highlighting their frequent involvement in AI research initiatives. In addition, disciplines such as social sciences and humanities, as well as environmental and physical sciences, are regularly mentioned, demonstrating a broad, cross-sector engagement with AI across various national contexts.

1.4. Oslo meeting

As part of this MLE, a dedicated meeting on Talent for AI in Science took place in Oslo, Norway, between 6 and 7 February 2025. The meeting gathered experts, policymakers, and researchers from across Europe to exchange insights, discuss challenges, and identify best practices related to AI talent development in science.

The meeting was attended (in-person or online) by 34 representatives from research funding agencies, ministries of education and research, universities, and research institutions from Austria, Flanders (Belgium), Denmark, Estonia, France, Germany, Ireland, Italy, the Netherlands, Norway, Slovenia, Sweden, and the United Kingdom. Experts from SINTEF Digital, the Research Council of Norway, NORA (Norwegian AI Network), Helmholtz Information and Data Science Academy (Germany), EMBL-EBI, and the University of Cambridge were among the key contributors.

The discussions revolved around major themes affecting AI talent in science, including:

- Education and training: Strategies for integrating AI into scientific curricula, best practices for interdisciplinary AI education, and mechanisms for upskilling and reskilling scientists.
- Careers and retention of talent: Challenges in retaining talent for AI in science in academia, the role of public-private partnerships in career development, and the increasing competition with industry for AI-skilled professionals.
- Interdisciplinarity in AI research: The need for better collaboration between AI experts and domain scientists, policy measures to foster interdisciplinary AI research, and the structural challenges in scientific publishing for interdisciplinary studies.
- Policy and funding initiatives: Examples of national strategies supporting AI talent, funding mechanisms for interdisciplinary AI research, and lessons from European AI clusters and research initiatives.

Insights from this meeting provided essential input for this report, complementing findings from the survey and desk research. The discussions highlighted the urgent need for structured policies to attract, train, and retain AI talent within scientific research and academia in Europe.

2. Talent attraction and retention for AI in science

2.1. Introduction

Attracting and keeping talented individuals is essential for advancing the use of AI in science. In this context, 'talent' encompasses diverse skill sets, including AI research; the application of existing AI tools to other scientific domains; research, software, and data engineering; data curation for scientific disciplines; and vital 'horizontal' competencies such as team-working, communication, and responsible innovation. With AI playing a growing role in solving complex scientific problems, there is an increasing need for **highly skilled experts** who can develop and apply AI tools effectively (in this sense, experts could be: AI/machine learning researchers, data scientists, data engineers, data stewards, scientific software developers, domain scientists, and similar). Unlike other hot fields (e.g. biomedical research), the competition for AI expertise is especially intense because AI's rapid pace of innovation and wide-ranging applicability attract strong interest across multiple sectors, from industry to government, amplifying the scramble for talent. The competition for such talent is fierce, as industries, academia, and governments worldwide seek to build strong AI capabilities[6].

Science faces unique challenges in attracting, developing, and retaining AI talent. Private companies often offer higher salaries, better resources, and faster career progression, making it difficult for research institutions and universities to compete for specialist AI talent. Additionally, the demand for interdisciplinary skills—combining expertise in AI with deep knowledge in specific scientific fields—further limits the available talent pool.

MLE countries face a range of challenges in retaining and attracting AI talent in the science sector. The most frequently cited issues include a lack of long-term funding opportunities and inadequate career development support, coupled with low salaries and more precarious work contracts compared to industry. Additionally, several responses highlighted difficulties in accessing state-of-

[6] For an overview of the over 1,000 AI policy initiatives launched by various countries, territories, and the EU, visit the OECD AI Policy Observatory (OECD, 2021).

the-art research infrastructure and limited interdisciplinary collaboration. In some cases, concerns were raised about gender disparities, restricted international mobility, and excessive bureaucracy, which further hinder the ability to advance scientific projects efficiently.

This section explores the challenges and opportunities in attracting and keeping AI talent in science. It looks at factors such as working conditions, career pathways, and the importance of creating supportive environments that encourage researchers to thrive and contribute to the advancement of AI-driven science.

2.2. Challenges and opportunities

Attracting and retaining talent in AI for science presents both challenges and opportunities. Key factors influencing this dynamic include working conditions, career pathways, long employment visa processes for non-EU citizens, and the creation of supportive environments that enable researchers to excel in AI-driven scientific fields.

Overall, while most initiatives are open to researchers across all scientific areas rather than specifically targeting AI or hybrid talent, there are notable exceptions. In the Netherlands, the ICAI is specifically designed to support AI talent, and in Germany, programmes under the Humboldt Foundation provide AI-focused opportunities. Italy has also introduced special PhD resources for AI using NextGenerationEU funds. In contrast, responses from Slovenia, Austria, Flanders (Belgium), Ireland, and several other entries indicate that initiatives generally do not specifically target AI or hybrid talent, with most national calls encompassing all areas of science rather than being tailored for AI research.

2.2.1. Challenges

2.2.1.1. Working conditions

The private sector often offers AI professionals higher salaries, better resources, and more rapid career advancement compared to academic and research institutions. This disparity makes it challenging for the scientific community to attract and retain top AI talent. For instance, industries provide access to proprietary datasets and advanced computing resources, which are sometimes lacking in academic settings. This advantage allows private companies to recruit AI specialists for diverse projects. Discussions at the Oslo meeting highlighted that private sector salaries and resources far exceed those available in academia, making it difficult for universities and research institutions to retain AI talent. Many AI researchers, especially those skilled in machine learning and deep learning, are drawn to industry due to higher pay, access to proprietary datasets, and advanced computing resources. The Norwegian AI Cloud representative and the NORA Network panellists noted that many PhD graduates in AI leave academia soon after completing their studies, with industry providing better-defined career trajectories and funding security.

Technology companies continue to dominate as employers of tech graduates, but the financial services sector is rapidly gaining ground, with Cornell University reporting a rise in computer science graduates entering finance from 16% in 2022 to 22% in 2023, and Carnegie Mellon noting a similar trend among its information systems graduates[7]. In our survey, only three MLE countries reported collecting official statistics on academic researchers transitioning between academia and the private sector. These are Ireland, Norway, and Flanders (Belgium). Flanders' Ecoom findings[8] show: "*The largest share of PhD holders leaves the Flemish university after obtaining their PhD:*

[7] Why the Coolest Job in Tech Might Actually Be in a Bank, The Wall Street Journal, 9 December 2024 (https://www.wsj.com/articles/why-the-coolest-job-in-tech-might-actually-be-in-a-bank-274330c5)

[8] https://www.ecoom.be/en/research/research_careers/nutshell

80% of PhD holders are no longer academically active at a Flemish university five years after obtaining their PhD (cohort PhD holders 2009-2012)."

While the transition of AI talent from academia to industry presents challenges, it is also a natural and beneficial outcome of a dynamic knowledge economy. Not all AI professionals need to stay in academia, and industry careers offer valuable pathways for knowledge application, technology transfer, and cross-sector collaboration. However, the rapid expansion of AI-related private sector opportunities has shifted the balance significantly, increasing the difficulty for research institutions to retain AI expertise.

Compounding this challenge is the lack of adequate infrastructure, particularly access to high-performance computing (HPC) resources. While initiatives like the Norwegian AI Cloud and EuroHPC are working to close this gap, many researchers still struggle with limited computational power and the high costs associated with advanced AI research, putting them at a disadvantage compared to industry, which offers far superior resources.

Lengthy administrative processes for hiring international AI researchers, securing grants, and setting up collaborations slow progress and create frustration. Another key issue is the lack of clear policies to retain AI talent in academia. Unlike industry, which offers structured career paths and attractive opportunities, universities often have no long-term plans to keep AI researchers, making it easy for them to leave for better-paying and more stable positions elsewhere.

Despite these challenges, **academia offers unique benefits that the private sector cannot always match. Many researchers value the intellectual freedom, long-term scientific exploration, and public good orientation of academic research. The ability to work on foundational AI research, contribute to open science, and pursue curiosity-driven inquiries** without immediate commercial pressures makes academia attractive to certain individuals. Ensuring that academic careers remain engaging, mission-driven, and well-supported is crucial for retaining AI talent in science.

2.2.1.2. Career pathways

The evolving nature of AI necessitates continuous learning and skill development. **Traditional academic career pathways may not offer the flexibility or opportunities for interdisciplinary collaboration** that AI professionals seek. To address this, some institutions have implemented **structured career progression frameworks** that assess contributions in research, teaching, and service to the academic community. These frameworks aim to create positive working environments and provide clear advancement opportunities[9]. Participants in the Oslo meeting highlighted that AI researchers in academia face significant career uncertainty due to the prevalence of temporary contracts, project-based funding, and limited tenure-track positions, making long-term planning difficult. Of the nine countries that responded to our survey, only Estonia and Germany have introduced flexible career paths or tenure-track opportunities for AI researchers.

AI research spans computer science, domain-specific fields, and data science, yet traditional academic career paths often fail to support this combination of skills. Many researchers struggle to build careers in science because **universities tend to favour specialisation over interdisciplinary work.** This challenge is made even more difficult by rigid evaluation criteria, as leading AI journals often prioritise theoretical advancements over practical applications. Scientists applying AI in fields such as medicine, physics, or climate science frequently find it difficult to get published, which limits their academic progression.

[9] Academic Career Pathways (Research and Teaching) (ACP R&T), University of Cambridge (https://www.acp.hr.admin.cam.ac.uk/)

Another major issue discussed in Oslo was the **difficulty of returning to academia after working in industry.** Strict hiring policies, outdated evaluation systems focused on academic publishing, and a lack of incentives for collaboration between industry and universities create significant barriers, making it hard for researchers to transition back.

2.2.2. Opportunities

Creating an environment that fosters innovation and collaboration is crucial for retaining AI talent in science. Encouraging teamwork and cross-functional collaboration enhances job satisfaction among AI professionals. For example, organisations that promote a collaborative culture can keep employees engaged and loyal, thereby reducing turnover[10].

Despite the challenges we have mentioned, there are significant opportunities to attract and retain AI talent in science:

Building strong partnerships between AI researchers and scientists from different fields can drive innovation and make academic research roles more attractive. Successful models, such as the NORA Network in Norway and Helmholtz Information and Data Science Academy (HIDA) in Germany, demonstrate the benefits of interdisciplinary AI training. These initiatives focus on integrating AI across various scientific disciplines, fostering collaboration between experts from different backgrounds, and creating a more dynamic and engaging research environment.

HIDA, for instance, organises postgraduate courses open to participants from any academic background—whether or not they have prior experience with AI. These courses cover topics such as data science, machine learning, programming, statistics, HPC, and data visualisation, aiming especially to support non-AI researchers in gaining relevant skills. In contrast, NORA functions as a PhD school in AI, with a mission to attract students from across disciplines to pursue doctoral studies in AI. Rather than catering to different profiles of AI researchers, these programs exemplify different strategies for spreading AI knowledge and building interdisciplinary bridges.

Collaboration between academia and industry play a crucial role in supporting AI researchers by giving them access to resources, industry expertise, and hands-on experience while maintaining connections with academia. Collaborations between universities and industry not only enhance research but also help professionals develop practical skills. Examples of successful initiatives include the Accelerate Programme at the University of Cambridge and the Norwegian AI Cloud, both of which enable researchers to work on real-world AI applications while staying engaged with academic institutions. Similarly, Germany's AI Centres of Excellence and Norway's RCN AI research funding programme actively promote AI-industry collaboration by offering joint funding opportunities, ensuring that researchers in both sectors benefit from shared knowledge and resources.

Creating flexible career pathways can help retain AI talent by allowing movement between academia, industry, and international organisations, giving researchers a wider range of experiences and opportunities. Initiatives such as the ARISE Career Accelerator at EMBL and Germany's Emmy Noether Programme provide early-career AI researchers with flexible funding and independent research positions, helping them establish themselves in the field. Some universities are also exploring hybrid roles, where researchers divide their time between academia and industry, ensuring better knowledge transfer and stronger collaboration between the two sectors.

Developing hybrid researcher roles and providing opportunities for domain upskilling can help bridge the gap between AI and scientific expertise. Specialised training programmes that

[10] AI talent acquisition and retention challenges and strategies, AiMaturity (https://aimaturity.ai/index.php/2024/10/21/ai-talent-acquisition-and-retention-challenges-and-strategies/)

equip domain experts with AI skills—and AI specialists with scientific knowledge—are essential for creating researchers who can work effectively across disciplines. Norway's national AI strategy includes initiatives to support these hybrid roles, ensuring that scientists can integrate AI into their work. Similarly, Germany's HIDA focuses on training PhD students and postdocs in both data science and domain-specific AI applications.

Supporting non-academic staff, such as data stewards, technical specialists, and AI-support personnel, is essential for maintaining a strong AI research ecosystem. These professionals play a key role in managing large datasets, maintaining research infrastructure, and ensuring that scientists can effectively apply AI in their work. The EMBL-EBI data training initiative is a great example of how training and developing these roles can enhance AI-driven research, making it more efficient and sustainable.

2.3. Policies for talent attraction and retention

2.3.1. Overview of existing policies in MLE participating countries

A range of funding initiatives support researcher mobility and the attraction of scientists across the surveyed MLE countries:

- In Slovenia, while there are few initiatives targeting mobility from the private sector post-PhD, several programmes support international mobility, including MSCA COFUND actions, the Dr. Aleš Debeljak Programme for returning researchers, the Slovenian Research and Innovation Agency's (ARIS) bilateral and international partnership grants, COST Actions, ERA-NET projects, and specific schemes such as ARIS "ERC focus" and Gravity projects.
- In Austria, measures include funding from the Austrian Science Fund (FWF), tax reductions for foreign researchers, and targeted recruiting efforts through OeAD[11] (Agency for Education and Internationalisation) and regional programmes like the Erwin Schrödinger Programme and Vienna Research Group initiatives.
- The Netherlands has established research funding initiatives that promote mobility, such as the Innovation Centre for AI (ICAI), AiNed Fellowship Grants, and attractive research funding programmes by NWO.
- In Flanders (Belgium), mobility instruments and specific grants for national and international researchers are in place[12].
- Ireland benefits from university-centred talent recruitment schemes.
- Italy supports international mobility through Industrial PhD Scholarships and significant tax breaks for researchers returning from abroad.
- In Germany, initiatives include programmes facilitated by the Humboldt Foundation, DAAD, and the previously existing AI Alexander von Humboldt Professorship.

Our survey indicates that only four countries—Germany, Ireland, the Netherlands, and Slovenia—have **specific funding programmes or policies to retain and attract AI talent in the academic sector**:

- In Germany, there are multiple funding options supporting AI researchers, including Alexander von Humboldt professorships in AI, funding for junior AI research groups, AI Centres of

[11] https://oead.at/en/
[12] https://www.fwo.be/en/support-programmes/international-mobility/

Excellence, and PostdocNetAI. A key focus of the Deutsche Forschungsgemeinschaft's (DFG) strategic funding initiative on AI is to attract and retain early-career AI experts, notably through independent junior research groups in the Emmy Noether Programme. These groups aim to recruit the next generation of highly qualified AI researchers by providing them with significant autonomy and funding at an early career stage. Additionally, Germany is also taking advantage of dedicated international recruitment initiatives.

- Research Ireland funding agency also offers various funding opportunities for academic and industry research, though they are not exclusively for AI.
- The Netherlands has the NGF AiNed Fellowship Grant[13] programme, which seeks to draw AI talent to Dutch academic research institutions, recognising the global competition for AI expertise. They have also established public-private partnerships for AI-related research, as in Austria, which additionally offers special funding for AI research positions.
- In 2024, Slovenia introduced two dedicated calls—Gravity and Strategic Projects—funding four major initiatives (2.25 million EUR each), two of which focus on AI (Language Models and AI for Science).

Many countries recognise the importance of AI in science, but funding for AI research is often too broad and not specifically aimed at developing and retaining AI talent. While some nations, such as Germany, the Netherlands, and Slovenia, have tried to introduce some AI-focused funding programmes, many others still treat AI as part of general research budgets, leaving researchers without the targeted support they need to advance their careers. On top of this, bureaucratic hurdles make things even harder.

2.3.2. Policy recommendations

Based on insights from the survey and discussions at the Oslo meeting, desk research and the author's expert view, several key areas require policy attention to effectively attract and retain talent for AI in science. To foster a more supportive and sustainable environment for AI talent, policymakers should focus on:

- Targeted AI research funding that specifically supports AI-focused career development, training, and interdisciplinary research.
- Streamlined administrative processes to reduce bureaucratic delays in hiring and funding applications.
- Simplified visa policies to foster international mobility and attract and retain global AI talents.
- Clear policies for talent retention, including structured career progression, tenure-track positions, and incentives (e.g. AI-focused awards or dedicated chairs) to encourage long-term engagement in academia.
- Enhance research salaries to increase their competitiveness versus the private sector and retain and attract AI and scientific talent.

Enhancing research salaries is a strategic necessity to retain top AI and hybrid and scientific talent in academia and should not be ignored. Below is a list of options that countries could explore:

[13] https://www.nwo.nl/en/researchprogrammes/national-growth-fund/ained/ained-fellowship-grants

- Establish a European-wide Research Salary Index to benchmark public-sector researcher salaries against industry equivalents, and implement annual reviews to align pay with inflation and market trends.
- Develop sustainable and flexible funding models, including dedicated government-backed salary funds, tax incentives for universities raising salaries, and co-financing arrangements with industry and EU programmes for AI collaborations.
- Improve researcher job stability and mobility, through long-term contracts with competitive pay for early- and mid-career researchers, fast-track mechanisms for salary adjustments, mobility-based pay to reflect cost-of-living differences, and market-based salary negotiation flexibility, especially in AI disciplines.

3. AI education and curricula integration

3.1. Introduction

Education is the foundation for building a strong pipeline of talent in AI for science. To meet the growing demand for AI expertise in scientific research, it is essential to integrate AI education into curricula from primary schools to university-level programmes. Early exposure to AI concepts helps foster interest and develop foundational skills; moreover, such initiatives can contribute to overcoming gender stereotypes in STEM fields by encouraging broader participation and engagement across diverse demographics. Advanced university programmes should then equip students with the specialised knowledge needed for interdisciplinary research.

Embedding AI education across all levels of learning ensures a larger and more diverse pool of talent, capable of addressing complex scientific challenges. By introducing computational thinking and ethical AI use early on, and offering tailored graduate programmes that combine AI with scientific disciplines, education systems can better prepare future generations to drive innovation in AI-powered science.

Overall, the ratings for the current level of AI education in schools and universities vary considerably across the surveyed MLE countries. In countries such as the Netherlands, some respondents rated the level as excellent—with AI education being well-integrated across various levels and fields—while others rated it as poor, citing its absence or inadequacy. Similarly, in Slovenia, Austria, and Ireland, responses tend to lean towards an average level, noting that AI education is limited to a few institutions or programmes. Meanwhile, Estonia, Flanders (Belgium), Italy, and several responses from Germany indicate that although AI education is growing, it still has room for improvement.

Discussions at the Oslo meeting reinforced the importance of strengthening AI education at all levels. Participants stressed that while some countries are making progress, many still lack structured AI curricula, particularly at the primary and secondary school levels. The need for coordinated strategies to integrate AI education systematically was highlighted as a priority for ensuring a well-prepared AI talent pipeline.

One of the key challenges is the rapid evolution of AI technologies, requiring education systems to be more agile and responsive. AI courses, especially at university and professional training levels, must be frequently updated to reflect emerging trends such as foundation models, generative AI, and ethical considerations. Without continuous course revisions, AI education risks becoming outdated, leaving researchers and professionals unprepared for real-world AI applications.

Integrating AI education into curricula from primary through university levels is crucial for cultivating a robust talent pool in AI for scientific research. This comprehensive approach ensures that students develop the necessary skills and knowledge to apply AI effectively across various scientific disciplines.

This section explores how AI education and curriculum integration can contribute to developing talent for AI in science.

3.2. Opportunities for integrating AI in education

3.2.1. Early exposure in primary and secondary education

Introducing AI concepts at an early age fosters interest and foundational understanding. For example, Singapore has piloted AI-integrated science lessons in secondary schools, helping students grasp AI's relevance in scientific contexts[14].

In Norway, AI-related concepts are introduced in primary school (ages 6–14), although AI is not formally part of the curriculum. Oslo meeting participants noted that while AI is not yet part of the national curriculum in Norway, initiatives such as the University of Bergen's digital understanding courses and public library-based AI literacy programs help fill the gap by strengthening AI education among students and the general public.

In the Netherlands and Estonia, AI is not yet formally taught, with the Netherlands planning to integrate it into the core objectives for digital literacy, and Estonia introducing such concepts in secondary school (ages 14–18). Conversely, in Ireland and Austria, AI is a mandatory part of the secondary school curriculum, while in Flanders (Belgium) pupils are exposed to algorithmic thinking in primary school without formal coverage of AI, data science, or machine learning.

In the Netherlands, efforts to revise educational curricula to incorporate AI more comprehensively are in the early stages; while AI is not yet part of the formal curriculum, it will soon be included in the core objectives for digital literacy, with formal enshrinement in law expected by 2027. Similarly, Slovenia, Estonia, Ireland, Italy, and Austria are all undergoing reforms to incorporate AI education—with Slovenia and Estonia introducing AI concepts in high school computer science courses, Ireland implementing the Digital Strategy for Schools to 2027, Italy initiating discussions and scattered projects in secondary schools, and Austria pursuing ongoing curricular reforms. Participants in the Oslo meeting highlighted that in Slovenia and Estonia, government-backed initiatives fund afterschool digital workshops, often hosted by universities and private sector organisations, to introduce children to AI outside formal curricula. In contrast, Norway and Flanders (Belgium) currently have not yet started to revise educational curricula to more comprehensively include AI.

3.2.2. Interdisciplinary university programmes

At the tertiary level, interdisciplinary programmes that combine AI with scientific fields prepare students for complex problem-solving. The Chinese University of Hong Kong's AI curriculum for pre-tertiary education exemplifies this by co-creating AI courses that bridge engineering and education, enhancing teachers' and students' AI literacy[15]. Digital Linguistics[16] is an international

[14] Park, J., Teo, T.W., Teo, A. et al (2023), Integrating artificial intelligence into science lessons: teachers' experiences and views. IJ STEM Ed 10, 61 (2023), (https://doi.org/10.1186/s40594-023-00454-3).

[15] T. K. F. Chiu, H. Meng, C. -S. Chai, I. King, S. Wong and Y. Yam (2022), "Creation and Evaluation of a Pretertiary Artificial Intelligence (AI) Curriculum," in IEEE Transactions on Education, vol. 65, no. 1, pp. 30-39, Feb. 2022, doi: 10.1109/TE.2021.3085878.

[16] https://digiling.university/

joint Master's programme offered by the University of Ljubljana, Masaryk University, and the University of Zagreb. The study programme is open to students with backgrounds in linguistics, computer science, or social science.

The Italian National PhD Programme in Artificial Intelligence[17] consists of five federated courses across 61 universities and research institutions, each targeting a strategic sector such as health, agrifood, government, industry, or society. It offers a shared foundational curriculum in AI—covering topics like multi-agent systems, machine learning, natural language processing, and robotics—and aims to train researchers and innovators to tackle complex, multidisciplinary challenges in the AI ecosystem.

During the Oslo meeting, participants underscored the importance of interdisciplinary AI education, with several examples highlighted:

- The NORA Network[18] in Norway enables students to take AI courses across multiple universities while earning credits at their home institutions, fostering interdisciplinary collaboration.
- The University of Bergen is developing a Master's programme in AI and Learning Sciences to address the growing demand for domain-specific AI training.
- Germany's HIDA[19] ensures that PhD-level interdisciplinary AI training is embedded into scientific disciplines, including life sciences and engineering.
- Slovenia's Data Science Master Programme[20] at the University of Ljubljana was mentioned as a model bridging AI expertise with other scientific fields.
- Italy's National PhD Programme in AI[21] provides an interdisciplinary AI curriculum, training researchers in AI for scientific discovery, health, and industry applications.

In the Netherlands and Slovenia, AI education is largely confined to computer science degrees rather than being part of the standard curriculum across other disciplines. In contrast, Estonia, Ireland, Italy, Austria, and Flanders (Belgium) incorporate AI-related subjects at both undergraduate and postgraduate levels across a range of fields—including life sciences, physical sciences, environmental sciences, engineering, and, in some cases, social sciences and humanities. In Austria, universities have the freedom to adapt their curricula to include AI, with current efforts aimed at boosting digital competencies for all students, while Flanders (Belgium) also offers AI education through specialised professional bachelor's programmes at technical institutions.

3.2.3. Professional development and lifelong learning

Ongoing training for educators and professionals ensures that AI knowledge remains current. Initiatives like Intel's "AI for Workforce" programme collaborate with community colleges to provide extensive AI content, facilitating continuous education and job readiness[22]. The University of Cambridge runs an Accelerate Programme for Scientific Discovery[23] offering workshops and so-called AI-clinics to train researchers in AI. In Germany, the ML Science Collaboratory[24] offers

[17] https://www.phd-ai.it/en/359-2/
[18] https://www.nora.ai/
[19] https://www.helmholtz-hida.de/en/
[20] https://datascience.fri.uni-lj.si/
[21] https://www.phd-ai.it/en/
[22] AP News, Intel exec on bringing artificial intelligence into the workplace (https://apnews.com/article/ai-workforce-intel-training-initiatives-26772553fe7ed82b535b41a8169ac078)
[23] https://science.ai.cam.ac.uk/events
[24] https://mlcolab.org/

practical workshops that demonstrate how science can benefit from machine learning, covering both introductory concepts and advanced, relevant ML techniques.

The Oslo meeting emphasized the role of public-private partnerships in AI education. The NORA Research School in Norway was highlighted as a strong model for industry-academia collaboration, offering flexible AI courses tailored to industry needs while keeping students engaged in research. Similarly, the Norwegian AI Cloud[25] provides training for researchers in AI infrastructure, ensuring they have access to HPC resources.

3.2.4. Addressing ethical and societal implications

Integrating AI ethics into education helps students understand the broader impact of AI technologies. For example, discussing algorithmic bias can illustrate how AI systems may unintentionally discriminate against certain social groups. More broadly, socio-economic impacts could include ethical dilemmas in autonomous vehicles, AI-generated content and (mis)information disorders, and the environmental footprint of digital technologies.

The World Economic Forum advocates for AI literacy that includes critical evaluation of AI's potential and limitations, promoting informed and responsible use[26].

3.3. Examples of educational initiatives integrating AI

Integrating AI education into curricula worldwide is essential for developing a skilled workforce capable of advancing scientific research and innovation. Various educational initiatives across different levels aim to equip students with AI knowledge and skills.

3.3.1. Primary and secondary education

A number of recent reports and projects are aimed at improving AI curricula offered from the primary school level upwards:

- UNESCO's Global Mapping of K-12 AI Curricula: UNESCO's report highlights that only 11 countries have developed and endorsed AI curricula for K-12 education, with an additional four countries in the process of development. This underscores the need for broader implementation of AI education at the primary and secondary levels to build foundational skills[27].
- AI4K12 Initiative (USA): The AI4K12 initiative in the United States is developing national guidelines for teaching AI in K-12 education, aiming to integrate AI concepts into early education to foster interest and understanding from a young age[28.]
- The Generation AI project[29] from Finland is launching an AI app along with educational resources for primary school children. The research-based initiative enables children to safely explore and understand AI technologies.

[25] https://www.naic.no/english/
[26] World Economic Forum (2024), How we can prepare for the future with foundational policy ideas for AI in education, (https://www.weforum.org/stories/2024/04/prepare-future-policy-ideas-ai-in-education/)
[27] UNESCO (2022), K-12 AI Curricula: A mapping of government-endorsed AI curricula, (https://unesdoc.unesco.org/ark:/48223/pf0000380602)
[28] Heintz, F. (2021), Three Interviews About K-12 AI Education in America, Europe, and Singapore. Künstl Intell 35, 233–237 (2021), (https://doi.org/10.1007/s13218-021-00730-w)
[29] https://www.generation-ai-stn.fi/

Several MLE countries have launched initiatives to encourage early AI education among children and teenagers. In the Netherlands, national programmes like NOLAI[30] are at the forefront; funded by the National Growth Fund, NOLAI develops intelligent technologies to enhance primary and secondary education while exploring the pedagogical and societal implications of smart tech. Similarly, Ireland has implemented a suite of national programmes, including the Digital Strategy for Schools to 2027 and various STEM and digital skills policies, aimed at embedding digital technologies and boosting coding competencies across the entire education system. Austria also has national programmes dedicated to early AI education.

Other countries rely on more localised or private-sector driven initiatives. In Slovenia and Estonia, government-supported calls encourage providers to offer afterschool workshops on digital competencies, including AI and computer science, often delivered by universities and private entities. Norway and Flanders (Belgium) follow a similar pattern, with initiatives supported outside the formal curricula—often as voluntary or extracurricular hobbies—offering children opportunities to engage with AI concepts through coding, robotics, and related activities.

The Oslo meeting discussions reinforced the need for structured AI curricula and better coordination between policymakers and educators to ensure AI education aligns with future workforce needs. Participants noted that while several MLE countries have promising initiatives, AI education is still fragmented, with many relying on extracurricular or industry-supported programmes rather than formal integration into curricula.

3.3.2. University-level programmes

In several MLE countries, universities have implemented specific cross-disciplinary programmes aimed at teaching AI to non-computer science students. In the Netherlands, for example, dedicated programmes and courses offer insights into AI, often highlighting its ethical implications, providing students from other disciplines with a solid foundation in the field. Ireland has introduced similar initiatives, such as the Higher Diploma in Science in Artificial Intelligence Applications, which are designed for students with diverse academic backgrounds. Italy offers several second-level master's programmes in Data Science and AI that target non-computer science graduates, such as the one available at the University of Pisa. In Austria, universities are actively expanding their curriculum to include AI education for all students; some institutions have already developed micro-credentials or AI-based training courses, while others are planning extensions in the coming years—ranging from AI engineering modules to workshops that explore AI literacy in the arts. Norway integrates AI concepts within its existing science courses, ensuring that students across various scientific disciplines gain exposure to the technology. Similarly, in Flanders (Belgium), dedicated cross-disciplinary programmes are available to fields as diverse as linguistics, medicine, and law, with specialised master's programmes that further deepen non-computer science students' understanding of computational linguistics, human language technology, and AI/ML concepts.

Across the MLE countries surveyed, there are some non-engineering and non-computer science university programmes that typically cover some foundational AI topics. In the Netherlands, courses include basic programming and coding skills, algorithms and data structures, machine learning, the ethics of AI, and AI applications in science and research. In Slovenia, topics focus on basic programming and coding, ethics, and key analytical skills such as statistics and data handling. Ireland and Italy offer a more comprehensive curriculum that also incorporates neural networks, deep learning, alongside basic programming, algorithms, machine learning, ethics, AI applications, and data handling skills. In Austria, general AI courses are designed to develop an understanding of how AI—particularly generative AI—functions, blending technical, ethical, and

[30] https://www.ru.nl/en/nolai

legal aspects to empower thoughtful and responsible use of AI. Meanwhile, in Norway, the emphasis is on machine learning, ethics, AI applications in science, and data analytics, and in Flanders (Belgium), courses focus on basic programming, AI applications, and analytical skills.

Oslo meeting participants highlighted the importance of dedicated cross-disciplinary AI programmes that allow non-computer science students to gain AI literacy. The University of Cambridge's Accelerate Programme and Germany's AI Centres of Excellence were cited as strong models for integrating AI into scientific education and training.

3.3.3. International collaborations and initiatives

As AI education becomes increasingly critical to scientific and technological progress, international collaborations and initiatives play a key role in shaping curricula, addressing skills gaps, and promoting early AI literacy. Various organisations and academic institutions are working to develop frameworks that support the integration of AI education across different levels of learning. Below are some initiatives that aim to enhance AI education through cross-border collaboration and policy alignment:

- TeachAI Initiative: Organisations Code.org, ETS, ISTE, Khan Academy, and the World Economic Forum have launched TeachAI to develop frameworks assisting governments and educational authorities in integrating AI into primary and secondary curricula worldwide, focusing on safe and ethical AI education[31].
- AI+ Initiative (Europe): Coordinated by the Universidade da Coruña, the AI+ initiative aims to introduce AI into the secondary school curriculum across Europe, promoting early engagement with AI concepts among students[32].
- ARISA seeks to address skills shortages and gaps by developing an AI Skills Strategy for Europe, aligning with evolving AI job requirements, and delivering curricula and learning programmes to equip individuals with essential AI knowledge and skills[33].

3.4. Policy recommendations

Based on the discussions at the MLE meeting in Oslo, desk research and the author's expert view, several recommendations were made to strengthen AI education and curriculum integration:

- Include topics from AI and computational thinking into primary and secondary schools.
- Expand university AI education beyond computer science to include students across various scientific disciplines.
- Develop flexible and modular AI courses to allow students from different backgrounds to acquire AI skills at various stages of education.
- Support lifelong learning and professional development in AI, particularly for domain researchers who need to integrate AI into their work.
- Strengthen public-private partnerships to create practical AI training opportunities, ensuring researchers gain hands-on experience with AI applications.

[31] TeachAI (https://www.teachai.org/)
[32] AI+ Initiative (https://aiplus.udc.es/)
[33] ARISA project (https://aiskills.eu/)

- Encourage cross-university collaboration, enabling students to access AI courses from multiple institutions.
- Incentivise scientists to engage with the public to demystify AI and promote informed discussions on its societal impact.
- Ensure agility and responsiveness of the educational system by regularly re-evaluating and updating course content. This could include periodic audits of AI curricula and modular learning structures that allow for quick adaptation to technological advancements.

4. Training and upskilling

4.1. Introduction

Training and upskilling are vital for developing and maintaining a strong talent pool in AI for science. As AI technologies evolve rapidly, researchers and professionals need continuous learning opportunities to stay updated with the latest tools and methods. Providing accessible and effective training programmes ensures that scientists can apply AI in innovative ways to tackle complex challenges.

Discussions at the Oslo meeting underscored that AI training is often inaccessible to non-computer scientists, with major barriers including limited time, motivation, and structured learning pathways. Addressing these challenges is crucial to ensuring that AI becomes a core tool across scientific disciplines. Participants stressed the importance of integrating AI training into funded research grants and career development plans, allowing researchers to systematically build AI expertise alongside their primary scientific work.

This section explores the importance of equipping individuals with the right skills and highlights strategies to enhance training and upskilling efforts in the field.

4.2. The importance of equipping individuals with the right skills

AI is transforming the way science is conducted, from analysing massive datasets to automating experiments. To fully harness its potential, researchers must have the skills to design, implement, and interpret AI models effectively. However, many scientists lack formal training in AI, while AI specialists may lack domain-specific knowledge. This skills gap can limit progress in applying AI to scientific discovery.

Upskilling ensures that scientists are not left behind as AI technologies evolve. Training programmes that combine AI techniques with domain expertise empower researchers to address complex challenges, such as climate modelling, drug discovery, and materials science. A skilled workforce is also essential for maintaining global competitiveness in AI-driven science[34].

[34] International Science Council (2024), A guide for policymakers: Evaluating rapidly developing technologies including AI, large language models and beyond, Paris, France, International Science Council. DOI: 10.24948/2024.07 (https://council.science/publications/policy-makers-guide-framework-digital-technologies/); OECD (2023), Artificial Intelligence in Science: Challenges, Opportunities and the Future of Research, OECD Publishing, Paris, (https://doi.org/10.1787/a8d820bd-en).

MLE countries Austria, Flanders (Belgium), Germany, Ireland, and the Netherlands have specific policies, funding programmes, or initiatives aimed at upskilling researchers from various scientific disciplines to use AI in their research.

Austria has launched several initiatives aimed at enhancing the AI capabilities of researchers across various scientific disciplines. A cornerstone of these efforts is the AIM AT 2030 national strategy, initiated by the Austrian government. This strategy focuses on fostering AI research and development by upskilling researchers through cross-disciplinary training and qualification programmes. AIM AT 2030 seeks to integrate AI knowledge into key sectors such as healthcare, energy, and education, while also supporting the creation of AI learning ecosystems and the development of related infrastructure.

Another significant initiative is Digital Austria, which complements the national strategy by promoting digital transformation and innovation across the country. The Austrian Research Promotion Agency (FFG) plays a crucial role by providing various funding schemes that support research and innovation, including specific programmes dedicated to AI applications in science. For instance, the "Artificial Intelligence in Business and Research" programme enables researchers to leverage AI technologies to advance their work effectively.

At the institutional level, Johannes Kepler Universität Linz (JKU) is actively involved in AI research through its Cluster of Excellence Programmes. A notable example is the “Bilateral AI” project, which unites researchers from multiple institutions to advance AI technologies and incorporate them into interdisciplinary research efforts. This project benefits from substantial funding, facilitating collaboration and the enhancement of AI skills among researchers from diverse fields.

The purpose of the DFG Research Training Groups is to develop early-career researchers through structured training and focused research. The programme is funded for up to nine years and places a strong emphasis on interdisciplinary approaches. It is designed to prepare doctoral researchers for a variety of scientific careers, equipping them with the skills and experiences needed for success in diverse fields. Examples of these groups include: GRK 2868: D^3 – Data-driven Design of Resilient Metamaterials[35], and GRK 2853: Neuroexplicit Models of Language, Vision, and Action[36].

Ireland has several key initiatives and research centres focusing on AI. One prominent example is the Research Ireland Centre for Research Training in Artificial Intelligence, which aims to develop skilled researchers in cutting-edge AI technologies[37]. In addition, Trinity College Dublin and the ADAPT Centre have established the AI Accountability Lab[38], dedicated to exploring issues of fairness, transparency, and accountability in AI. Several Research Ireland Centres also conduct fundamental research and offer training in AI and related fields:

- ADAPT – AI-driven digital content technology
- INSIGHT – Data analytics
- Lero – Software research
- EMPOWER – Data governance

During the Oslo meeting, participants noted that AI training efforts often focus on PhD students and early-career researchers, while mid-career and senior researchers are frequently overlooked. Given the increasing role of AI in scientific research, it is essential to expand training opportunities to all career stages. Germany’s HIDA was highlighted as an example of a programme that offers

[35] https://tu-dresden.de/ing/forschung/graduiertenkollegs/grk2868?set_language=en
[36] https://www.neuroexplicit.org/
[37] https://www.sfi.ie/funding/centres-research-training/artificial-intelligence/
[38] https://www.adaptcentre.ie/news-and-events/trinity-college-dublin-and-adapt-establish-ai-accountability-lab/

AI training for researchers at multiple levels, ensuring that AI expertise permeates the academic community beyond early-career scientists.

4.3. Key strategies to enhance training and upskilling

4.3.1. Integrating AI into curricula

Educational institutions play a pivotal role in preparing the next generation of scientists with AI skills. Universities should embed AI and machine learning courses into STEM curricula at both undergraduate and postgraduate levels. These courses should not only cover fundamental AI/ML principles and methods but also emerging topics, such as effective strategies for querying LLMs for research purposes[39]. Examples include programmes that teach AI in combination with biology, physics, or chemistry, fostering interdisciplinary expertise. Such initiatives align with recommendations in the OECD's report on AI in science[40].

For instance, from 2025 to 2027, Austrian public universities have entered into performance agreements with the ministry to prioritise AI upskilling and reskilling initiatives for both students and higher education teachers and scientists. These universities are expected to develop new training programmes or enhance existing ones by 2027 to achieve these goals. It is important to note that these initiatives will be funded through the universities' overall budgets, rather than through separate funding channels, ensuring that AI education and training are integrated seamlessly into their existing financial frameworks.

Participants at the Oslo meeting emphasised that AI should not be seen as a separate discipline but rather as a fundamental tool applicable across scientific domains. To address this, they recommended integrating AI components into PhD programmes and postdoctoral training, ensuring that future researchers across disciplines gain foundational AI knowledge.

More on integration of AI into curricula has been discussed in section 3.

4.3.2. Providing lifelong learning opportunities

Continuous professional development is critical as AI technologies evolve. Online courses, workshops, and certification programmes can help scientists update their skills without leaving their roles. Platforms like Coursera and edX already offer AI-related courses tailored for working professionals, and institutions should consider collaborations to create custom training content for scientists[41].

One of the primary challenges identified in the Oslo meeting was that researchers struggle to allocate time for AI upskilling. Participants recommended that AI training should be explicitly included as an eligible cost in research grants, ensuring that scientists have protected time for learning AI methodologies. In some cases, research grants could allow funding for AI training as a specific budget item.

Some countries have already introduced national programmes to help scientists gain AI skills. In Austria, the AIM AT 2030 strategy[42] places a strong focus on AI training to encourage its adoption

[39] AI4Science, M. R., & Quantum, M. A. (2023). The impact of large language models on scientific discovery: a preliminary study using GPT-4. *arXiv preprint arXiv:2311.07361*.

[40] OECD (2023), Artificial Intelligence in Science: Challenges, Opportunities and the Future of Research, OECD Publishing, Paris, (https://doi.org/10.1787/a8d820bd-en).

[41] OECD (2023), Artificial Intelligence in Science: Challenges, Opportunities and the Future of Research, OECD Publishing, Paris, (https://doi.org/10.1787/a8d820bd-en).

[42] https://www.digitalaustria.gv.at/eng/strategy/strategy-AI-AIM-AT-2030.html

in research The Austrian Science Fund (FWF) supports training and development programmes aimed at reskilling researchers in AI. This includes initiatives where researchers in traditional fields, such as social sciences or natural sciences, are trained in AI techniques to enhance their research capacities. France's National AI Strategy[43] provides dedicated funding to support scientists in upskilling, ensuring they can effectively integrate AI into their work. Meanwhile, in the Netherlands, the ICAI brings universities and industry together to offer specialised AI training for experts in various fields, helping bridge the gap between research and real-world applications.

4.3.3. Collaboration between academia and industry

Collaborations between academic institutions, governments, and private companies can provide access to cutting-edge resources and expertise. For example, programmes where industry experts co-deliver training with academics can bridge the gap between theory and practice. Such partnerships also open pathways for researchers to gain experience with state-of-the-art AI tools and datasets[44].

- Technology Ireland ICT Skillnet[45] is a government-backed business support network that connects teams and individuals from both small and large companies to highly subsidised and cutting-edge training in the latest technologies driving business development. They uniquely co-develop their courses with industry and academic partners, covering areas such as AI, Software Development, Cybersecurity, Data Analytics, Cloud, Fintech, and Digital Transformation. These courses are delivered in various formats to suit different learning preferences, including online and part-time master's programmes, on-demand self-paced industry-certified micro-credentials, and bespoke in-company training.
- VAIA accelerates the adoption of AI in Flemish companies by consolidating and distributing a comprehensive range of AI training courses for professionals and researchers. It assists training organisations in customising their programmes to address current and future AI demands by providing insights into the training needs of various professional sectors and connecting them with AI expertise from both academia and industry. Acting as a bridge-builder, VAIA facilitates the seamless exchange of AI knowledge between academics and businesses, thereby promoting lifelong learning in Flanders and ensuring that AI knowledge is accessible to interested parties. In addition to VAIA, intermediary professional organisations such as AGORIA[46] offer different types of courses including AI and are not limited to a specific industry.
- The Norwegian AI Cloud was recognised at the Oslo meeting as a successful model that not only provides computational infrastructure but also training for researchers to effectively use AI tools in their work. The initiative ensures that domain experts receive technical support alongside access to computing resources.
- Slovenia's MSCA Co-fund programme SMASH (Machine Learning for Science and Humanities Postdoctoral Programme)[47] was highlighted as a good practice in training researchers through industry placements, ensuring that scientists gain hands-on experience with AI applications in real-world settings.

[43] https://minefi.hosting.augure.com/Augure_Minefi/r/ContenuEnLigne/Download?id=334FD34F-7844-497E-9551-79EDFF3B2EEF&filename=1645%20-%20DP%20-%20Strat%C3%A9gie%20Nationale%20pour%20l%27IA%202%C3%A8me%20phase.pdf

[44] Alan Paic (2019), Policies for Artificial Intelligence in Science and Innovation, Artificial Intelligence for Science, Industry and Society, AISIS2019 October 21-25, 2019

[45] https://ictskillnet.ie/course-category/artificial-intelligence/

[46] https://www.agoria.be/en

[47] https://smash.ung.si/

4.3.4. Fostering interdisciplinary training

AI in science requires not just technical knowledge but also the ability to apply it across diverse fields. Training programmes should encourage interdisciplinary learning, combining AI techniques with problem-solving in areas like healthcare, environmental science, or engineering. Multidisciplinary PhD programmes or joint degrees in AI and domain sciences can serve this purpose[48].

Slovenia's University of Nova Gorica has been awarded the MSCA Co-fund project SMASH, co-funded by the Ministry of Higher Education, Science and Innovation. SMASH is an innovative, interdisciplinary career development programme designed to train outstanding postdoctoral researchers from around the world in advanced machine learning applications for science and the humanities. By supporting 50 exceptional postdocs, the project aims to enhance their career prospects while strengthening Europe's human capital in research and innovation.

Germany's AI Centres of Excellence[49] could serve as a model for integrating AI training into domain-specific research. These centres focus on training non-CS researchers in AI methodologies tailored to their scientific fields, ensuring that AI is seamlessly embedded into disciplines such as health sciences, engineering, and environmental research.

Participants at the Oslo meeting also noted that mentorship and peer-learning models play a crucial role in AI upskilling. Programmes like EMBL's ARISE Career Accelerator[50] were cited as effective structures that provide mentorship and structured career development in research infrastructure.

4.3.5. Ensuring inclusivity in training

Efforts to train and upskill must ensure inclusivity, addressing barriers faced by underrepresented groups in AI and science. Initiatives like scholarships for women in AI or outreach programmes for underserved regions can diversify the talent pool and bring varied perspectives to scientific research.

One major barrier to AI upskilling, discussed at the Oslo meeting, is that academic career structures do not sufficiently reward interdisciplinary AI training. Participants suggested that funding agencies and universities should include AI training and interdisciplinary research as formal evaluation criteria for tenure, promotions, and research grant applications. This would incentivise domain researchers to integrate AI training into their career progression rather than viewing it as an optional skill.

4.4. Policy recommendations

These policy recommendations are based on insights gathered from the MLE meeting in Oslo, survey responses from participating countries, desk research on best practices in AI training and upskilling, and the author's expert view. They address key challenges in ensuring that scientists

[48] International Science Council (2024), A guide for policymakers: Evaluating rapidly developing technologies including AI, large language models and beyond, Paris, France, International Science Council. DOI: 10.24948/2024.07 (https://council.science/publications/policy-makers-guide-framework-digital-technologies/); OECD (2023), Artificial Intelligence in Science: Challenges, Opportunities and the Future of Research, OECD Publishing, Paris, (https://doi.org/10.1787/a8d820bd-en).

[49] https://ki-zentren.net/en/

[50] https://www.embl.org/about/info/arise/

across disciplines can effectively integrate AI into their research and teaching while maintaining access to the necessary resources and support structures:

- Reform evaluation criteria to recognise AI training efforts as valuable (yet not mandatory) when performing research and teaching performance assessments.
- Introduce train-the-trainer programmes and faculty exchanges with AI-specialised institutions.
- Expand national AI infrastructures (e.g. Norwegian AI Cloud, Germany's AI Centres of Excellence) to provide both computing power and structured training.
- Introduce grants and subsidies for AI courses, similar to Austria's Digital Austria initiative, which funds AI training programmes for researchers.
- National and European research funding agencies should embed AI training costs in research grants.

5. Interdisciplinary AI research and career development

5.1. Introduction

Interdisciplinary research is essential for advancing AI in science, as it enables the integration of AI tools and techniques into fields like biology, physics, and environmental science. Such collaboration can lead to breakthroughs in solving complex scientific problems. However, achieving this requires deliberate efforts to foster interdisciplinary approaches and create sustainable career pathways that encourage scientists to engage across fields. At the same time, it must be acknowledged that interdisciplinary research is inherently challenging and may not always succeed[51].

Discussions at the Oslo meeting underscored that interdisciplinary AI research is often constrained by structural barriers within academic institutions. Participants highlighted that researchers working at the intersection of AI and domain sciences frequently encounter obstacles such as a lack of formal recognition for interdisciplinary work, difficulties in securing cross-disciplinary funding, and limited opportunities for AI training tailored to non-computer scientists. Addressing these challenges is essential to ensuring that AI is seamlessly integrated into scientific discovery.

Good initiatives to foster interdisciplinary research include creating dedicated funding programmes for projects that bring together AI experts and domain scientists, establishing interdisciplinary research centres, and offering joint academic programmes that combine AI with scientific disciplines. Career development can be supported by recognising and rewarding interdisciplinary work in academic and institutional settings, providing mentorship opportunities, and offering flexible career paths that bridge academia, industry, and public research organisations. This section explores these initiatives and strategies, focusing on how they can strengthen interdisciplinary research and career development for AI in science.

[51] MacLeod, M. (2018), What makes interdisciplinarity difficult? Some consequences of domain specificity in interdisciplinary practice. Synthese, 195(2), 697-720;
Abbonato, D., Bianchini, S., Gargiulo, F., & Venturini, T. (2024). Interdisciplinary research in artificial intelligence: Lessons from COVID-19. Quantitative Science Studies, 5(4), 922-935.

5.2. How to foster interdisciplinary AI research

Interdisciplinary AI research brings together diverse expertise to tackle complex scientific challenges. To foster such collaboration effectively, institutions and policymakers need to implement strategies that encourage the convergence of AI with other disciplines.

Across the surveyed MLE countries, there is considerable variation in programmes aimed at fostering organisational or institutional incentives for interdisciplinary AI research. While Slovenia and Ireland reported no specific initiatives in this area, Austria, Estonia, the Netherlands, Flanders (Belgium), Italy, and Germany have implemented several measures. For example, Austria has introduced bilateral and hybrid AI funding calls, Comet Centres, AI factories, and university-level initiatives such as CAIML and GRAML at TU Vienna, as well as national calls on topics like hybrid AI and AI for Green. Estonia supports interdisciplinary AI research through the Estonian Centre of Excellence in Artificial Intelligence. In the Netherlands, interdisciplinary research is promoted via initiatives like the ICAI, the University of Amsterdam's AI4Science Lab, interdisciplinary labs at TU Delft, and TU Eindhoven's cross-departmental AI institute (EAISI). In Flanders (Belgium) there is a range of initiatives, including Flanders AI Research, Leuven.AI, Delta at the University of Ghent, the Antwerp Centre for Responsible Artificial Intelligence, AI@UHasselt, and FAIR – AI for the common good. Italy has recently established AI4Industry, an institute focusing on automotive, aerospace and AI research, while Germany utilises transdisciplinary research areas, the Cluster of Excellence 'Science of Intelligence', and DFG Research Units in AI to support interdisciplinary collaboration.

At the Oslo meeting, participants noted that despite the presence of these initiatives, AI researchers and domain scientists often work in separate academic environments, making collaboration difficult. Institutional silos prevent effective knowledge exchange, and many researchers struggle to integrate AI into their work due to a lack of targeted training programmes.

Overall, most AI scientists across the surveyed countries believe that there is insufficient funding specifically dedicated to interdisciplinary AI research at the national level. In Slovenia, funding schemes are not explicitly targeted at research that bridges AI with other scientific disciplines, and the general national funding calls are too broad and topic-specific to address foundational interdisciplinary needs. In Austria and Estonia, there is a consensus that while some applied research areas like GreenAI are funded, there is a significant gap in supporting the interdisciplinary aspects of AI, especially as the field becomes increasingly multimodal and complex. Similarly, in the Netherlands and Flanders (Belgium), available funding is either too narrowly focused on applications or is struggling to keep pace with the growing costs and technological demands, resulting in limited support for fundamental interdisciplinary AI research. Responses from Ireland and Italy also indicate that AI funding tends to be siloed by discipline or hindered by sector-specific career paths, while opinions in Germany are mixed—some programmes provide support for interdisciplinary initiatives, but many respondents feel that additional funding is needed to foster a broader diversity of approaches in mainstream AI research.

Oslo meeting participants emphasised that national research funding is often structured around traditional disciplines, making it difficult for AI researchers and domain experts to jointly apply for grants. Funding schemes tend to prioritise discipline-specific projects, leaving interdisciplinary research at a disadvantage when competing for financial support.

Some actionable approaches supported by examples of good practices are provided below.

5.2.1. Provide funding for collaborative projects

Funding programmes that specifically require interdisciplinary teams can incentivise collaboration. Examples include:

- The European Union's Horizon Europe Framework, which funds projects that integrate AI with domains like environmental sustainability and health[52].
- The US National Science Foundation's (NSF) AI Institutes, which support interdisciplinary partnerships to advance AI in areas such as agriculture and manufacturing[53].
- The Eric and Wendy Schmidt AI in Science Postdoctoral Fellowship at Oxford[54] is an international initiative to promote innovative AI use in STEM research. Over six years, Oxford will host approximately 55 postdoctoral fellows, providing them with AI and machine learning tools to accelerate and broaden their research. Imperial's I-X Centre for AI in Science[55] is launching a new Global Fellowship programme, supported by Schmidt Sciences, to empower researchers from Ghana, wider Africa, India, Brazil, and Malaysia to enhance their scientific research with cutting-edge AI. The programme, announced alongside Imperial Global Ghana, its first science and technology hub in Africa, will support three cohorts of fellows over two years, enabling them to co-locate at I-X and build local capacity, with funding equivalent to 48 years of research time.
- The DFG provides Priority Programmes funding opportunities that can support AI in Science. The goal of the Priority Programmes is to provide coordinated, multi-location funding for innovative research topics that significantly advance science and the humanities. The programme focuses on topics with the potential to shape entire research fields, particularly encouraging interdisciplinary proposals. It promotes collaboration among different research locations to foster networking and provides support for early-career researchers involved in the projects. Additionally, the programme includes tailored equity measures to ensure inclusive and diverse participation. Examples of these programmes include: SPP 2298: Theoretical Foundations of Deep Learning[56], SPP 2353: Daring More Intelligence – Design Assistants in Mechanics and Dynamics[57], and SPP 2331: Machine Learning in Chemical Engineering[58].

Participants at the Oslo meeting highlighted that while these international funding initiatives exist, **national funding structures often do not prioritise interdisciplinary AI research**. As a result, researchers in many MLE countries struggle to secure financial backing for cross-disciplinary projects, particularly in fundamental research areas where AI applications may not yet be well established. Participants recommended that national research funding agencies **explicitly allocate funding for interdisciplinary AI research and provide dedicated support for researchers transitioning into AI from other fields**. Without targeted financial incentives, interdisciplinary AI research is unlikely to reach its full potential.

5.2.2. Develop joint academic programmes

Interdisciplinary education lays the foundation for future collaboration. Joint degree programmes and training initiatives can combine AI and domain-specific expertise:

[52] OECD (2023), Artificial Intelligence in Science: Challenges, Opportunities and the Future of Research, OECD Publishing, Paris, (https://doi.org/10.1787/a8d820bd-en).
[53] International Science Council (2024), A guide for policymakers: Evaluating rapidly developing technologies including AI, large language models and beyond, Paris, France, International Science Council. DOI: 10.24948/2024.07 (https://council.science/publications/policy-makers-guide-framework-digital-technologies/)
[54] https://saiis.web.ox.ac.uk/
[55] https://www.imperial.ac.uk/ix-ai-in-science/
[56] https://www.foundationsofdl.de/
[57] https://www.itm.uni-stuttgart.de/en/spp_2353/
[58] https://chemengml.org/

- The University of Cambridge provides a Computational Biology programme, integrating AI and bioinformatics training to solve challenges in genomics and personalised medicine.
- The UKRI Artificial Intelligence Centres for Doctoral Training[59] are set to train a new generation of PhD students who will develop novel AI methodologies and apply AI technology in areas such as healthcare, climate change, and commercial innovation. Following a £100 million investment in 2018, UKRI further invested £117 million in 12 new centres in 2023—training around 900 students over eight years through partnerships with 370 organisations across the private, public, and third sectors.
- European PhD and post-doc initiatives in computational sciences such as post-doctoral programme MSCA COFUND SMASH[60] bridge AI with fields like environmental modelling, physics, linguistics, medicine and similar.
- Several Austrian academic institutions, such as the University of Vienna, Vienna University of Technology (TU Wien), and the Austrian Research Institute for Artificial Intelligence (OFAI), offer specific courses, workshops, and specialised training programmes. These programmes are designed to help researchers from various disciplines gain hands-on experience with AI technologies. Some of these courses are part of their PhD programmes or short-term offerings focused on AI applications in different domains.

The Oslo meeting underscored that the integration of AI into PhD and postdoctoral programmes is crucial for fostering interdisciplinary AI research. However, many universities still treat AI as a separate discipline, rather than embedding AI training into existing scientific curricula. Participants recommended expanding joint AI programmes that combine AI expertise with domain-specific knowledge.

5.2.3. Incentivise collaboration through career rewards

Interdisciplinary work often requires additional effort, such as bridging gaps in knowledge or aligning diverse research goals. Institutions can:

- Recognise interdisciplinary publications and projects during tenure and promotion reviews.
- Establish awards for interdisciplinary research achievements, as seen in programmes like the Royal Society's Research Professorships[61], which supports scientists working at the intersection of fields.

At the Oslo meeting, participants highlighted that research evaluation systems continue to favour discipline-specific achievements rather than recognising AI applications across multiple fields. This creates a disincentive for researchers to engage in interdisciplinary AI work. To address this, participants recommended that universities and funding agencies revise evaluation criteria to account for contributions that span AI and domain sciences.

5.2.4. Use shared data and tools to facilitate collaboration

Interdisciplinary research often relies on access to shared datasets, tools, and computational resources:

[59] https://www.ukri.org/who-we-are/our-vision-and-strategy/tomorrows-technologies/how-we-work-in-ai/ukri-artificial-intelligence-centres-for-doctoral-training/
[60] MSCA COFUND project SMASH: machine learning for science and humanities postdoctoral programme
[61] Royal Society Research Professorship (https://royalsociety.org/grants/research-professorship/)

- The Human Cell Atlas Project uses open data and AI to map human cells, bringing together biologists, computational scientists, and clinicians[62].
- Federated Data Platforms: Europe's GAIA-X Initiative promotes shared, secure data infrastructure to support interdisciplinary projects across sectors[63].
- The European Open Science Cloud (EOSC): Designed to provide researchers with access to data and AI tools from various fields[64].

5.2.5. Establish dedicated interdisciplinary research centres

Creating specialised centres for interdisciplinary AI research enables sustained collaboration among experts from diverse fields. For example:

- The Alan Turing Institute in the UK focuses on AI and data science research, promoting partnerships between computer scientists, statisticians, and domain experts from fields like medicine and climate science[65].
- Germany's Cyber Valley Initiative combines expertise in AI and robotics with applications in materials science and healthcare, facilitating cross-disciplinary innovation[66].
- Austria has several competence centres, such as the Austrian Institute of Technology (AIT), which provide researchers with advanced AI training through hands-on projects and collaborations with industry partners. These centres are essential for enabling researchers from various scientific domains to learn how to incorporate AI into their work.

5.3. Career development challenges

Developing a career in AI for science poses unique challenges, particularly because of the interdisciplinary nature of the field. To foster collaboration across various scientific domains, it is essential to create new career paths such as hybrid research positions, data managers, data stewards, and software research engineers. In the Netherlands, policy initiatives support these roles, although not all are specifically targeted at AI-enabled research. Slovenia is addressing this need by introducing data steward positions through national legislation and the nationally funded project "Spoznaj," which provides extensive training focused on open science with some emphasis on AI aspects in data management. Meanwhile, in Ireland, the Higher Education Authority funded key initiatives in 2022 to advance Open Research, culminating in the establishment of the national data stewardship network, Sonraí[67]. Sonraí aims to develop data stewardship skills across the national research landscape, enhance the professionalisation of data steward roles, and support the FAIR data principles, ensuring the effective curation, preservation, and dissemination of national data assets. These initiatives collectively aim to strengthen interdisciplinary AI-enabled

[62] OECD (2023), Artificial Intelligence in Science: Challenges, Opportunities and the Future of Research, OECD Publishing, Paris, (https://doi.org/10.1787/a8d820bd-en).
[63] Paic, A. (2019), Policies for Artificial Intelligence in Science and Innovation, Artificial Intelligence for Science, Industry and Society, AISIS2019 October 21-25, 2019
[64] Paic, A. (2019), Policies for Artificial Intelligence in Science and Innovation, Artificial Intelligence for Science, Industry and Society, AISIS2019 October 21-25, 2019
[65] OECD (2023), Artificial Intelligence in Science: Challenges, Opportunities and the Future of Research, OECD Publishing, Paris, (https://doi.org/10.1787/a8d820bd-en).
[66] International Science Council (2024), A guide for policymakers: Evaluating rapidly developing technologies including AI, large language models and beyond, Paris, France, International Science Council. DOI: 10.24948/2024.07 (https://council.science/publications/policy-makers-guide-framework-digital-technologies/)
[67] https://datastewards.ie/

research by building the necessary professional infrastructure and enhancing the skills of researchers across different scientific fields.

Oslo meeting participants emphasised that interdisciplinary AI career pathways are often unclear, with few structured opportunities to transition between AI and domain sciences. Researchers working at this intersection frequently struggle to establish credibility in both AI and their primary scientific field, which can negatively impact their career progression.

Several key career development challenges faced by AI professionals in scientific research are discussed below:

Lack of recognition for interdisciplinary work

Many academic and research institutions continue to prioritise traditional disciplinary achievements over interdisciplinary contributions. Researchers working at the intersection of AI and other scientific fields often find their work undervalued when applying for promotions or grants[68]. Oslo meeting participants noted that this creates a career disadvantage for researchers who integrate AI into their work but do not fit neatly into a single academic discipline.

Unclear career progression pathways

Career trajectories for AI researchers in science and for research support personnel are often ambiguous. Since the field spans academia, industry, and public research, researchers face uncertainty in choosing and navigating their professional paths. There is also a lack of standardised benchmarks for evaluating interdisciplinary careers[69].

Across all the surveyed countries, career support for AI researchers working on interdisciplinary projects is generally not provided at a national level. Instead, such support tends to be left to individual institutions or to the researchers themselves, with no specific national programmes dedicated to this aspect of career development.

Participants at the Oslo meeting stressed the need for clearer career frameworks that support transitions between AI and scientific disciplines, ensuring that researchers have viable long-term career options.

Insufficient mentorship

AI in science is a relatively new field, and there are limited senior researchers or mentors with experience in both AI and its scientific applications. This gap leaves early-career researchers with little guidance on how to navigate their careers in this multidisciplinary space[70].

Difficulty in securing long-term positions

AI scientists often work on short-term projects or grants, making it difficult to secure stable, long-term positions in academia or research institutions. MLE participants noted that this challenge is especially pronounced in interdisciplinary roles, where researchers may struggle to secure permanent positions within traditional department structures. This insecurity discourages many from pursuing academic careers.

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[68] OECD (2023), Artificial Intelligence in Science: Challenges, Opportunities and the Future of Research, OECD Publishing, Paris, (https://doi.org/10.1787/a8d820bd-en).

[69] OECD (2023), Artificial Intelligence in Science: Challenges, Opportunities and the Future of Research, OECD Publishing, Paris, (https://doi.org/10.1787/a8d820bd-en); Paic, A. (2019), Policies for Artificial Intelligence in Science and Innovation, Artificial Intelligence for Science, Industry and Society, AISIS2019 October 21-25, 2019.

[70] International Science Council (2024), A guide for policymakers: Evaluating rapidly developing technologies including AI, large language models and beyond, Paris, France, International Science Council. DOI: 10.24948/2024.07 (https://council.science/publications/policy-makers-guide-framework-digital-technologies/)

Balancing academia and industry demand

AI talent is in high demand across industries, which often offer better pay and resources than academia. Researchers in academia may struggle to resist industry opportunities, particularly when their contributions to interdisciplinary research are not adequately rewarded[71].

Limited funding for interdisciplinary research

Funding bodies often allocate resources to well-defined disciplines, while interdisciplinary projects involving AI and science are viewed as higher risk. This lack of targeted funding creates challenges for researchers who need to establish credibility across multiple fields[72].

In Austria, the FFG offers specific funding schemes and support for research projects that integrate AI into diverse fields. For example, their programmes often include educational components where researchers can participate in AI training to apply AI methods in areas such as healthcare, business, and engineering.

Challenges in publication and peer recognition

Interdisciplinary work often faces difficulties in publication, as it may not align with the traditional focus of discipline-specific journals. Furthermore, peer reviewers from one field may not fully understand or appreciate the methods and contributions of another[73].

Geographic and institutional disparities

Researchers in developing countries or less-resourced institutions often lack access to the advanced infrastructure, funding, and networking opportunities required for impactful interdisciplinary AI research[74].

5.4. Policy recommendations

The following policy recommendations are based on discussions at the MLE meeting in Oslo, survey responses, desk research, and the author's expert view. They aim to address structural challenges in interdisciplinary AI research and career development by improving funding mechanisms, adapting evaluation criteria, and ensuring that AI becomes an integral part of national research and education strategies.

- National research funding agencies should introduce interdisciplinary AI funding schemes.
- Funding calls should explicitly require collaboration between AI researchers and domain experts.
- Evaluation processes should be adapted to include interdisciplinary expertise in review panels.
- Universities should embed AI across curricula to promote cross-disciplinary learning.

[71] European Commission, Directorate-General for Research and Innovation (2023), AI in Science: Harnessing the power of AI to accelerate discovery and foster innovation, 2023, doi:10.2777/401605

[72] OECD (2023), Artificial Intelligence in Science: Challenges, Opportunities and the Future of Research, OECD Publishing, Paris, (https://doi.org/10.1787/a8d820bd-en); Paic, A. (2019), Policies for Artificial Intelligence in Science and Innovation, Artificial Intelligence for Science, Industry and Society, AISIS2019 October 21-25, 2019.

[73] OECD (2023), Artificial Intelligence in Science: Challenges, Opportunities and the Future of Research, OECD Publishing, Paris, (https://doi.org/10.1787/a8d820bd-en).

[74] European Commission, Directorate-General for Research and Innovation (2023), AI in Science: Harnessing the power of AI to accelerate discovery and foster innovation, 2023, doi:10.2777/401605.

- AI training should be integrated into STEM and social science disciplines to build AI literacy across domains.
- PhD and postdoctoral programmes should support interdisciplinary AI research from the start. Early-career researchers should be paired with mentors from both AI and domain sciences.
- Develop clearer academic and industry pathways that encourage movement between fields without penalties.
- AI research should be embedded in national science strategies. Governments should ensure AI is a core component of national research agendas, requiring collaboration across fields.
- The evaluation of interdisciplinary projects should account for the inherent challenges of cross-field integration and tolerate failure, particularly in the early stages.

6. Public-private partnerships and international collaboration

6.1. Introduction

Public-private partnerships and international collaboration are vital for developing and retaining talent for AI in science. The challenges of advancing AI in science, such as access to resources, funding, and cutting-edge expertise, require cooperation across sectors and borders. Partnerships between academia, industry, and governments can bridge resource gaps, provide real-world experience, and create new opportunities for talent development. Similarly, international collaboration fosters the exchange of knowledge, access to global research infrastructure, and the pooling of talent to tackle complex scientific challenges. This section explores how these partnerships and collaborations can support talent development and strengthen AI's role in science.

The Oslo meeting highlighted that such collaborations are not only beneficial but also essential for ensuring AI expertise is applied effectively across scientific disciplines. Several speakers emphasized the need for stronger engagement between universities and the private sector, particularly in providing industry-relevant training, developing AI infrastructure, and supporting long-term career opportunities for AI researchers. Norway's NORA Network and the Norwegian AI Cloud were presented as strong models of public-private cooperation. Furthermore, participants discussed the role of international collaboration in bridging skills gaps and enhancing mobility for AI talent.

6.2. Examples of successful public-private partnerships and international collaboration

Public-private partnerships and international collaborations have played a crucial role in AI talent development, knowledge-sharing, and technological advancements. Below are examples of successful initiatives that demonstrate how these collaborations can enhance AI's role in science by providing training, research opportunities, and infrastructure for scientists and AI professionals.

The Alan Turing Institute (UK) - Public-private partnership

The Alan Turing Institute collaborates with universities, industry partners, and government agencies to advance AI research and its application across various domains, including healthcare, climate science, and engineering.

Talent Impact: It offers fellowship programmes, industry placements, and collaborative research opportunities for AI scientists, fostering a blend of academic and practical expertise.

The European Open Science Cloud (EOSC) - Public-public and international collaboration

EOSC is an initiative of the European Commission to create a federated data ecosystem for researchers across disciplines and countries.

Talent Impact: By providing access to shared data and tools, EOSC enables scientists and AI professionals to collaborate effectively, driving interdisciplinary research and fostering international AI talent development.

Experts at the Oslo meeting stressed that open data infrastructures like EOSC are critical for AI-driven research, particularly in enabling cross-border collaboration and ensuring equitable access to AI tools.

CERN and AI Collaborations - International collaboration

CERN integrates AI into particle physics research and collaborates with institutions worldwide to develop machine learning tools for data analysis.

Talent Impact: Its international network allows researchers from diverse disciplines to apply AI in fundamental science, providing unparalleled learning and growth opportunities for AI talent.

Discussions at Oslo pointed out that CERN's collaborative AI efforts could serve as a model for fostering interdisciplinary AI research across scientific fields. Similar approaches are being explored in Norway's AI research ecosystem.

Google AI for Social Good - Public-private collaboration

Google partners with non-profits and academic institutions to use AI in solving global challenges such as health, conservation, and disaster response.

Talent Impact: The initiative provides AI researchers with resources, training, and opportunities to work on socially impactful projects, encouraging ethical AI practices.

The meeting in Oslo emphasized the importance of structuring such collaborations in ways that balance academic openness with industry innovation, ensuring that AI research benefits society broadly rather than being monopolised by private entities.

AI for Earth (Microsoft) - Public-private collaboration

Microsoft's AI for Earth programme supports researchers and organisations in applying AI to environmental sustainability challenges, including biodiversity conservation and climate modelling.

Talent Impact: By providing funding, AI tools, and cloud resources, it empowers scientists and researchers to integrate AI into their work, expanding their technical skills and career opportunities.

Climate Change AI - International collaboration

This global initiative unites researchers, industry professionals, and policymakers to use AI in addressing climate change.

Talent Impact: It promotes interdisciplinary research, provides networking opportunities, and supports AI talent in engaging with real-world applications.

The meeting participants stressed the need for national funding agencies to support similar interdisciplinary and international collaborations, particularly in AI for sustainability, to ensure knowledge-sharing across borders.

Innovation Centre for Artificial Intelligence (ICAI) - Public-private collaboration

ICAI aims to keep the Netherlands at the forefront of AI knowledge and talent development by creating and nurturing a national AI ecosystem. By sustainably managing resources from its activities, ICAI seeks to activate and enhance AI capabilities throughout the country.

Talent Impact: The initiative boosts AI talent in the Netherlands by attracting skilled professionals, providing diverse problems and data for them to work on, and supporting the growth of local and national AI ecosystems.

Future Artificial Intelligence Research Foundation (FAIR) - Public-private collaboration

The FAIR is a non-profit organisation dedicated to implementing AI initiatives funded under Italy's National Recovery and Resilience Plan. Operating through a Hub & Spoke model, FAIR collaborates with national research institutions, universities, and industry partners to advance frontier AI research, foster innovation, create and retain AI talent, and develop trustworthy and sustainable AI technologies.

Talent impact: FAIR aims to create, retain, and attract AI talent in Italy by fostering an inclusive and interdisciplinary AI ecosystem, thereby enhancing the development and sustainability of skilled professionals within the national AI landscape.

Norwegian Open AI Lab (NAIL) - Public-private collaboration

NAIL is a central hub for AI research, education, and innovation, hosted by the Faculty of Information Technology and Electrical Engineering at NTNU in Trondheim. NAIL conducts a wide range of AI projects, fosters strong partnerships between academia, businesses, and the public sector, and disseminates AI knowledge through various events and activities to a diverse audience.

Talent impact: NAIL strengthens AI talent in Norway by providing research opportunities, educational programmes, and collaborative partnerships that enhance the skills and expertise of students, researchers, and industry professionals.

Experts at the meeting emphasized the **importance of ensuring that such academies and training programmes remain accessible to scientists across disciplines**, not just to those with computer science backgrounds. Norway's performance agreement contract, which funds AI courses at public universities, was identified as a complementary policy model to expand AI training initiatives.

6.3. Policy recommendations

The following recommendations are based on insights from the MLE meeting in Oslo, survey findings, desk research, and the author's expert view. They aim to strengthen public-private and international collaborations in AI research and talent development by improving funding mechanisms, fostering long-term partnerships, and ensuring equitable access to AI resources.

- Governments should actively promote structured collaboration between academic institutions, private companies, and research organisations to develop AI talent and research capacity. This could include co-funded AI research projects, industry-sponsored PhD programmes, and AI training initiatives led by both academia and industry.
- Ensure AI training initiatives in public-private collaborations remain accessible to all disciplines.
- National and EU-level research funding agencies should introduce grants that explicitly support cross-border AI research partnerships and interdisciplinary AI applications. Similar to Horizon Europe's AI funding streams, these should facilitate knowledge-sharing and mobility for AI talent.

- Governments should invest in AI research infrastructure, such as HPC resources, cloud platforms, and open-access datasets, ensuring that academic and industry researchers have the necessary tools to collaborate effectively. Initiatives like the Norwegian AI Cloud can serve as a model for other countries.
- Policymakers should ensure that AI is embedded as a core component of national research strategies, requiring collaboration across different scientific disciplines and between public and private sectors. This could include national AI centres that facilitate partnerships between universities and industry.
- Tax incentives, subsidies, and funding programmes should encourage private companies to invest in AI research collaborations with universities, co-develop training initiatives, and provide industry placements for AI researchers.
- AI talent benefits from diverse experiences across sectors. Governments should establish mobility programmes that allow researchers to work across universities, private companies, and international research institutions, ensuring cross-sectoral knowledge transfer.
- National and international policies should promote open science principles by encouraging companies and research institutions to share AI tools, datasets, and research outputs. This fosters greater collaboration and accelerates AI advancements in science.
- Performance-based funding agreements between governments and universities (such as those in Austria and Norway) should include requirements for AI training initiatives, ensuring that higher education institutions prioritise AI upskilling across disciplines.
- Many AI research collaborations are currently tied to short-term projects with limited impact. Governments and funding bodies should encourage multi-year AI partnerships between academia and industry to provide stability and sustained investment in AI research and workforce development.

7. Conclusion

The development, attraction, and retention of talent for AI in science are crucial for maintaining Europe's competitiveness in research and innovation. This thematic report has explored key areas essential for fostering AI expertise in scientific disciplines, including education, training, interdisciplinary research, career development, public-private partnerships, and international collaboration. Through the analysis of survey responses, desk research, and insights gathered from the MLE meeting in Oslo, several challenges and opportunities have been identified, alongside actionable policy recommendations.

A key finding is that AI in science requires not only technical expertise but also the ability to integrate AI tools within specific domains. However, traditional academic structures and funding mechanisms often do not support interdisciplinary collaboration, making it difficult for researchers to fully leverage AI. To address these barriers, national policies must focus on integrating AI education into curricula, developing flexible upskilling initiatives, and fostering cross-sector collaboration between academia and industry. Additionally, structured career pathways and funding opportunities tailored to interdisciplinary AI research must be expanded.

Education and training remain the foundation for AI talent development. While some MLE countries have made strides in embedding AI education at various levels, many still lack systematic integration, particularly at primary and secondary school levels. Efforts should be made to ensure AI literacy across disciplines, with a focus on ethical usage of AI, computational thinking, and interdisciplinary problem-solving. Moreover, lifelong learning opportunities, including train-the-trainer programmes and modular AI courses, should be promoted to ensure that researchers across all career stages can acquire necessary AI skills.

Interdisciplinary research is critical for applying AI in scientific discovery, yet significant challenges persist. Institutional silos, a lack of recognition for interdisciplinary work, and insufficient funding mechanisms limit collaboration between AI specialists and domain experts. Policies should incentivise cross-disciplinary research, create dedicated funding streams for interdisciplinary AI projects, and adapt research evaluation criteria to better acknowledge contributions at the intersection of AI and science.

Public-private partnerships and international collaborations play a crucial role in strengthening AI talent for science. Successful models, such as Norway's NORA Network, the Netherlands' ICAI, and Germany's AI Centres of Excellence, demonstrate the value of industry-academic collaboration. Expanding such initiatives can provide access to computational resources, real-world AI applications, and structured training programmes that bridge academic and industry needs.

Finally, national and European policymakers must take a proactive approach to ensuring that AI talent remains in scientific research rather than migrating to other sectors. Measures such as streamlined administrative processes, competitive salaries in academia, and policies that facilitate mobility and retention should be prioritised. Additionally, targeted funding for AI in science, embedded within national AI strategies, will help secure Europe's position as a leader in AI-driven research.

In conclusion, AI talent is a strategic necessity for scientific progress. A coordinated effort across governments, research institutions, and industry is required to build an ecosystem where AI talent can thrive. By implementing the recommendations outlined in this report, European countries can create a sustainable and competitive AI workforce capable of addressing the scientific challenges of the future.

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Interdisciplinary AI research and career development

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The MLE on AI in science Topic 3: Talent for AI in Science thematic report explores the challenges and opportunities in developing, attracting, and retaining AI talent in science across Europe. It highlights the growing demand for AI expertise and the need for better education, training, and career pathways. AI must be integrated into curricula at all levels, while upskilling and lifelong learning are essential to keep pace with advancements. Interdisciplinary AI research faces barriers such as rigid academic structures, funding gaps, and lack of recognition. Public-private partnerships and international collaborations play a key role in fostering AI talent. However, low salaries and unstable careers in academia drive experts away, leading to a brain drain. Policy recommendations focus on improving funding, structured career progression, and stronger AI education policies. Embedding AI in national research strategies and evaluation frameworks is crucial. A coordinated effort from governments, universities, and industry is needed to build a sustainable and competitive AI workforce.

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